

API Guide

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This guide provides instructions and samples for getting started with the Valcre API. If you have any questions, please reach out to us at support@valcre.com.

Resources

Swagger API Documentation

Our Swagger API site includes detailed information about the API endpoints, including parameters, sample calls, and sample responses. You can also make API calls directly from this site after entering your authentication token, which you can get following the steps in [Authentication](#).

The Swagger API site is available at <https://api-core.valcre.com/swagger/>.

Data Dictionary

Our Data Dictionary provides additional details about our data fields that are available with the API, including samples and available values for dropdown fields.

The latest Data Dictionary can be downloaded at <https://go.valcre.com/data-dictionary>.

Headers

As with all https calls, headers should be set correctly for the API to function properly. Some headers will be set automatically by your API framework and tooling. Please specify the following for successful calls:

None

Content-Type: `application/json`

Authentication

To use the Valcre API you must include an authentication token that can be retrieved following the steps below.

Initial Authentication

To authenticate you will need the **username** and **password** of a registered Valcre user. You will also need the **client_id** and **client_secret** which you can request from your Valcre representative.

None

Request:

POST <https://auth.valcre.com/oauth/token>

Headers:

Content-Type: application/json

Body:

```
{
  "grant_type": "password",
  "client_id": "<client_id>",
  "client_secret": "<client_secret>",
  "username": "<login>",
  "password": "<password>",
  "scope": "offline_access",
  "audience": "https://valcre.api.com"
}
```

Refresh Tokens

The original authentication token you receive will expire within 24 hours, and will need to be refreshed. As part of the original authentication response you will also receive a **refresh_token** that can be used to easily re-authenticate and receive a new token.

To re-authenticate with your **refresh_token** and receive a new token:

None

Request:

POST <https://auth.valcre.com/oauth/token>

Headers:

Content-Type: application/json

Body:

```
{  
  { "grant_type", "refresh_token" },  
  { "refresh_token", <refresh token> }  
}
```

Sample Application

Our Sample Application provides examples for authenticating and making common API calls. The project file includes .NET source code that can be run in Visual Studio.

After running the application, you can sign in to your Valcre account on the Sign In tab.

Many of the queries reference specific records by their ID, and you'll want to update that ID for your own records. For example, the "Get property by ID" query will need to have Endpoint = "Properties(**25167**)" changed to "Properties(**23111125**)". You can get the record IDs from the end of the URL in our web app.

The latest sample project can be downloaded at <https://go.valcre.com/api-sample>.

The screenshot shows the 'Valcre API Sample' application window. It has a 'Requests' tab and a 'Sign In' button. The 'Samples' dropdown is set to 'Get jobs due within a week'. The 'Endpoint' is 'Jobs', '\$orderby' is 'DueDate', and '\$top' is '25'. The '\$filter' is '(Status eq 'Open' or Status eq 'Draft') and (DueDate gt 2024-08-13) and (DueDate lt 2024-08-21)'. The '\$select' is 'Id,Number,Name,Status,DueDate'. The '\$expand' is 'Owner(\$select=Id,DisplayName),Property(\$select=Id,Name,AddressStreet,AddressCity,AddressState),Client(\$select=Id,FirstName,LastName,Company)'. The 'Request' field shows the full URL: `https://api-core.valcre.com/api/v1/Jobs/?$orderby=DueDate&$top=25&$filter=(Status eq 'Open' or Status eq 'Draft') and (DueDate gt 2024-08-13) and (DueDate lt 2024-08-21)&$select=Id,Number,Name,Status,DueDate&$expand=Owner($select=Id,DisplayName),Property($select=Id,Name,AddressStreet,AddressCity,AddressState),Client($select=Id,FirstName,LastName,Company)`. The 'Method' is 'GET'. The 'Data' field is empty. The 'Response' field shows a JSON array of job objects:

```
{
  "@odata.context": "https://api-core.valcre.com/api/v1/$metadata#Jobs",
  "value": [
    {
      "Id": 236931,
      "Number": "SC-22006",
      "Name": "Burger King",
      "Status": "Open",
      "DueDate": "2024-08-14",
      "Owner": {
        "Id": 5208,
        "DisplayName": "Ross Phelps"
      },
      "Property": {
        "Id": 23462742,
        "Name": "Burger King",
        "AddressStreet": "1540 N Eastern Ave",
        "AddressCity": "Los Angeles",
        "AddressState": "CA"
      },
      "Client": {
        "Id": 13428027,
        "Company": "ABC Bank",
        "FirstName": "James",
        "LastName": "Johnson"
      }
    },
    {
      "Id": 245173,
      "Number": "SC-22009",

```


Sample API Requests

Get a Property record by its ID:

None

```
GET https://api-core.valcre.com/api/v1/Properties(<property id>)
```

Update a Property record with the specified data:

None

```
PATCH https://api-core.valcre.com/api/v1/Properties(<property id>)
```

```
{  
  "Name": "New name value",  
  "AddressStreet": "New address value",  
}
```

Create a Property record with the specified data:

None

```
POST https://api-core.valcre.com/api/v1/Properties
```

```
{  
  "Name": "Name value",  
  "AddressStreet": "Address value",  
}
```